

tell pyplot which colormap to use with the `cmap` argument. This code colors the points with lower *y*-values light blue and the points with higher *y*-values dark blue. Figure 15-8 shows the resulting plot.

NOTE

You can see all the colormaps available in *pyplot* at <https://matplotlib.org>. Go to *Tutorials*, scroll down to *Colors*, and click **Choosing Colormaps in Matplotlib**.

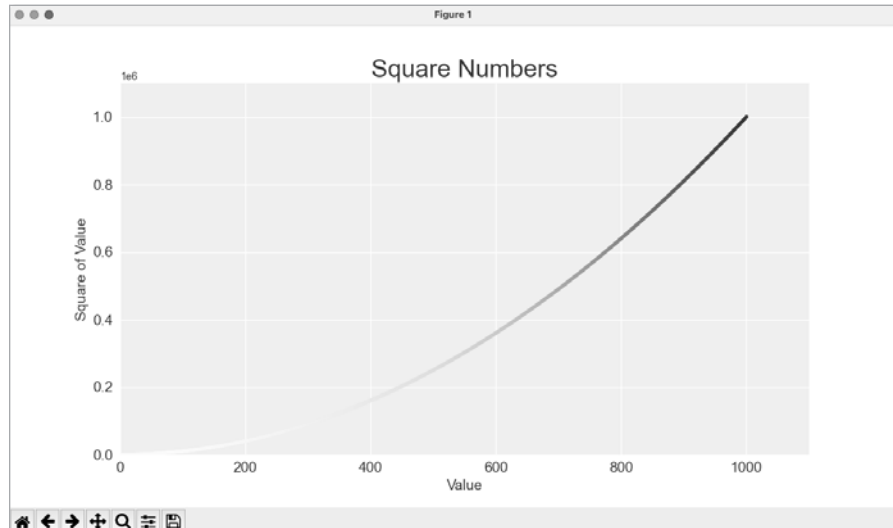


Figure 15-8: A plot using the *Blues* colormap

Saving Your Plots Automatically

If you want to save the plot to a file instead of showing it in the Matplotlib viewer, you can use `plt.savefig()` instead of `plt.show()`:

```
plt.savefig('squares_plot.png', bbox_inches='tight')
```

The first argument is a filename for the plot image, which will be saved in the same directory as *scatter_squares.py*. The second argument trims extra whitespace from the plot. If you want the extra whitespace around the plot, you can omit this argument. You can also call `savefig()` with a *Path* object, and write the output file anywhere you want on your system.

TRY IT YOURSELF

15-1. Cubes: A number raised to the third power is a *cube*. Plot the first five cubic numbers, and then plot the first 5,000 cubic numbers.

15-2. Colored Cubes: Apply a colormap to your cubes plot.